

HOSPITAL MANAGEMENT SYSTEM (HMS)

PROJECT PROPOSAL DOCUMENT – ADVANCED VERSION

1. PROPOSAL SUMMARY

We propose to develop an **Advanced Hospital Management System (HMS)** with a professional website, smart automation, WhatsApp integration, and AI-based face recognition for staff attendance. This solution is ideal for **mid-to-large hospitals (10-30 doctors)** and includes advanced reporting, digital prescriptions, IPD management, and comprehensive analytics.

Ideal For: Mid to large hospitals, multi-specialty centers, nursing homes (10-30 doctors)

2. OBJECTIVES

- Fully digitize all hospital workflows — OPD, IPD, and pharmacy
- Automate appointment, billing, and staff management
- Improve patient experience and reduce waiting time
- Implement smart attendance using face recognition
- Ensure secure, centralized, and audit-ready data handling

3. SCOPE OF WORK

3.1 Website Development (Advanced)

- Modern UI/UX design with hospital branding
- Doctor profiles with real-time availability
- Online appointment booking with slot management
- Patient login portal
- Health blog and hospital news section

Cost: Rs. 40,000 - Rs. 60,000

3.2 Doctor Appointment System (Smart)

- Real-time time-slot booking per doctor
- Auto doctor availability update
- Appointment rescheduling and cancellation
- Department-wise appointment view
- Calendar and history tracking

Cost: Rs. 60,000 - Rs. 80,000

3.3 Patient Management (Advanced OPD + IPD Ready)

- Patient registration with unique UHID
- Full visit history and timeline view
- Digital prescriptions and e-notes
- Lab report and document upload
- IPD admission and discharge records
- Advanced search and patient filters

Cost: Rs. 70,000 - Rs. 1,00,000

3.4 Staff Management + Face Recognition Attendance

- Staff profile and role management
- AI face recognition-based attendance system
- Real-time attendance tracking dashboard
- Shift and leave management
- Staff performance summary

Cost: Rs. 80,000 - Rs. 1,20,000

3.5 Medicine Inventory (Advanced)

- Detailed stock tracking with batch and expiry
- Supplier order management
- Auto low stock alerts
- Medicine usage and wastage tracking

Cost: Rs. 50,000 - Rs. 70,000

3.6 Billing & Revenue Management

- OPD and IPD billing generation
- GST-compliant invoice
- Payment history and due tracking
- Revenue reports by doctor/department

Cost: Rs. 40,000 - Rs. 60,000

3.7 Admin Dashboard (Advanced Analytics)

- Real-time hospital performance overview
- Patient, appointment, and revenue analytics
- Staff performance and attendance insights
- Custom report export (PDF/Excel)

Cost: Rs. 40,000 - Rs. 60,000

4. ADDITIONAL FEATURES

4.1 SMS + Email + WhatsApp Notifications

- Appointment confirmations and reminders
- Discharge summaries to patients via WhatsApp
- Lab report ready alerts

Cost: Rs. 18,000 - Rs. 28,000

4.2 Online Lab Report Management

- Lab test request and upload portal
- Patient-wise report history
- Digital report download by patient

Cost: Rs. 20,000 - Rs. 32,000

4.3 Role-Based Access Control (Enhanced)

- Multi-level roles (Admin/Doctor/Nurse/Pharmacist)
- Secure JWT authentication

Cost: Rs. 15,000 - Rs. 25,000

4.4 Cloud Backup & Data Security

- Daily automated cloud backup
- Secure encrypted storage

Cost: Rs. 15,000 - Rs. 20,000

4.5 Audit Logs & Activity Tracking

- Full system activity tracking
- Security and compliance monitoring

Cost: Rs. 15,000 - Rs. 25,000

5. TECHNOLOGY STACK (MERN + AI)

- **Frontend:** React.js
- **Backend:** Node.js (Express.js)
- **Database:** MongoDB
- **AI Integration:** OpenCV / AWS Rekognition (Face Recognition)
- **Hosting:** AWS Cloud

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 4,00,000 - Rs. 5,50,000 (One-time)
Infrastructure	Hosting: Rs. 4,000 - Rs. 10,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year

Maintenance Cost

Rs. 25,000 - Rs. 35,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	1 Week
Development	6-8 Weeks
Testing	1-2 Weeks
Deployment	1 Week
Total Duration	8-10 Weeks

8. LIMITATIONS

- No mobile app (can be added later)
- Limited insurance and TPA billing
- No telemedicine/video consultation

9. FUTURE UPGRADES

- Mobile application (Patient + Doctor)
- AI-based patient analytics
- Telemedicine and video consultation
- Full billing and insurance/TPA system

10. SUPPORT AND WARRANTY

- 1 month free support
- Bug fixing included
- Staff training provided

11. TERMS AND CONDITIONS

- 50% advance payment
- 50% after completion
- Additional features charged separately

12. CONCLUSION

This Advanced HMS solution provides a **powerful, scalable, and semi-automated system** with modern technologies like face recognition, WhatsApp integration, and advanced analytics — making it ideal for hospitals aiming for complete digital transformation.

13. CONTACT DETAILS

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